

R65 Help Booklet

Contents

- [animate](#)
- [arglib](#)
- [compile](#)
- [dir](#)
- [filelib](#)
- [graph](#)
- [graphics](#)
- [help](#)
- [iotest](#)
- [keys](#)
- [ladders](#)
- [ledlib](#)
- [mathlib](#)
- [nroff](#)
- [paint](#)
- [pcodes](#)
- [pedit](#)
- [plotlib](#)
- [R65](#)
- [ralib](#)
- [striolib](#)
- [strlib](#)
- [syslib](#)
- [table](#)
- [teklib](#)
- [timelib](#)
- [view](#)
- [wildlib](#)

animate

ANIMATE

```
func expaint:boolean;  
  Handles toggle graphics.  
  Provides code to paint and apply motion.  
  Returns true if animation loop should be  
  Stops if motion detects stop  
  (e.g. expaint:=false)
```

```
func exkey(ch:char):boolean;  
  Provides code to handle keys pressed.  
  Returns true if animation loop should be  
  stopped because of key pressed.  
  (e.g. exkey:=ch=ord(0);)
```

Usage:
{I IANIMATE:P}

```
animate(autorepeat)  
  Run the animation loop.  
  Set autorepeat to true if cursor key repetition  
  should start immediately, not after short  
  delay.
```

arglib

LIBRARY ARGLIB;

MEMORY

```
NUMARG    = $005f: integer&;
ARGLIST   = $0060: array[10] of integer;
ARGLISTS  = $0060: array[63] of char&;
ARGTYPE   = $00a0: array[31] of char&;
FILFLG    = $00da: integer&;
FILDRV    = $00dc: integer&;
FILCYC    = $0311: integer&;
FILCY1    = $0330: integer&;
FILNAM    = 0301: array[15] of char&;
FILNM1    = $0320: array[15] of char&;
FILSTP    = $0312: char&;
```

VARIABLES

```
_carg: integer;
```

ROUTINES

```
proc _argerror(e: integer);
proc _agetval(var value:integer;
              var default:boolean);
proc _agetstring(var string: array[15] of char;
                 var default: boolean;
                 var cyclus, drive: integer);
```

compile

COMPILE name[.cy[,drv]] [/x]

name: file name of program to compile

x: p: no hard copy print

l: put line info in object code

r: range checking for index

n: no loader file

Default drive is 1.

Compiler directives:

{ \$I filename } include file

{ \$R- } { \$R+ } range checking, default false

{ \$U- } { \$U+ } unique identifiers in scope

dir

DIR [drive] [name] [/s]

where	/s	Sort directory
	drive	Drive number, default 1
	name	name of a disk

Examples	DIR	DIR /S
	DIR 0	DIR 0 /S
	DIR PSOURCE	DIR PSOURCE /S

filelib

LIBRARY FILELIB;

MEMORY

```
FILFLG = $00da: integer&;  
FILDRV = $00dc: integer&;  
FILTYP = $0300: char&;  
FILNAM = $0301: array[15] of char&;  
FILCYC = $0311: integer&;  
FILSTP = $0312: char&;  
FILNM1 = $0320: array[15] of char&;  
FILCY1 = $0330: integer&;
```

ROUTINES

```
func _uppercase(ch1: char): char;  
proc _asetfile(name: array[15] of char;  
    cyclus,device: integer; subtype: char);  
func _freedrv(drive:integer; printit:boolean);
```

graph

GRAPH filename

display the graph of function values
stored in table on internal display

TGRAPH filename

display the graph of function values
stored in table on Tektronix 4010

MAKETABLE filename

make a table for graph/tgraph

graphics

LOW RESOLUTION GRAPHICS ON THE R65 COMPUTER

Graphics can be displayed in split mode in a graphic overlay, or in fullscreen mode.

To toggle between full screen and split mode use the key "ctrl-l".

To clear the graphics canvas and release the graphics memory use the command "GREND".

help

HELP topic

where topic is a command name or another
file with extention :H on drive 1 or 0.
Lists available topics if topic not found.

iostat

```
PROGRAM IOTEST
USES SYSLIB, STRLIB
VAR S1: CPNT
    FOUT: FILE
BEGIN
    WRITELN ('IOTEST')
    STRFIO('IOTEST:B',1)
    OPENW(FOUT)
    WRITELN(@FOUT, 'IOTEST')
    CLOSE(FOUT)
END.
```

keys

IMPORTANT KEYBOARD KEYS ON THE R65 SYSTEM

CTRL-l	split/full screen graphics
CTRL-r	print all on
CTRL-t	print all off
CTRL-<return>	clear to end of screen
CTRL-<right>	insert char
CTRL-<left>	delete char
CTRL-<down>	scroll down
CTRL-<up>	scroll up
SHIFT-CTRL	is required on Mac/UTM/Debian

ladders

LADDERS [/D]
game with ladders

/D endless demo mode

ledlib

LIBRARY LEDLIB;

MEMORY

LEDREG=\$1432: array[7] of char&;

ROUTINES

proc _ledstring(s:cpnt);

proc _ledstop;

proc _ledhex(d,p,digits: integer);

proc _ledbyte(d: integer);

mathlib

LIBRARY MATHLIB;

CONSTANTS

```
PI = 3.14159;  
E  = 2.71828;
```

ROUTINES

```
func fabs(x:real):real;  
func sqrt(n:real):real;  
func cos(x:real):real;  
func sin(x:real):real;  
func tan(x:real):real;  
proc writeflo(f:file;r:real);  
proc writef0(f:file;d:integer;  
    r:real;fl:integer;centered:boolean);  
proc writefix(f:file;d:integer;r:real);  
func readflo(f:file):real;  
func ln(r:real):real;  
func exp(x:real):real;  
func log(x:real):real;
```

nroff

NROFF(1)

NAME

=====

NROFF – text formatter for R65

SYNOPSIS

=====

NROFF *filnam[:sub[.drv]]* [*linewidth*]
Default line width is 40.

DESCRIPTION

=====

NROFF formats plain text files using simple dot commands. Lines starting with a dot are formatting requests. All other lines are normal text and are formatted using fill mode.

TEXT MODES

=====

Fill mode Words are wrapped to fit the current line width.
No fill mode Text is printed exactly as entered.
.fi Enable fill mode.
.nf Enable no fill mode.

HEADINGS

=====

.TH Manual header line.
.SH Section heading. Uses underline or inverse video depending on width.
.SS Subsection heading.

PARAGRAPHS

=====

.PP Start new paragraph.
.br Line break.
.sp [*n*] Insert blank lines.

INDENTATION

=====

.RS Increase indentation.
 .RE Restore indentation.
 .IP label Indented paragraph. Label followed by
 text with hanging indent.

TABS

=====

.ta n1 n2 ...Set tab stops.
 Tabs are expanded during output.
 If no tabs are defined, default spacing is 8
 cols.

TEXT STYLE

=====

.B text Bold line. On 40 columns inverse video
 is used.

INPUT RULES

=====

Maximum input line length depends on buffer
 size.
 Control characters except TAB are ignored.

OUTPUT WIDTH

=====

Line width may be given as optional parameter.
 Example NROFF MANUAL:B,1 80
 Widths above 40 use printer style headings.

TABLES

=====

Tables are created using no fill mode and tabs.
 Example:

COL1	COL2	COL3
====	====	====
AA	BBB	CCCC

LIMITATIONS

=====

No macros or registers.
 No escape sequences.
 Formatting is single pass.

AUTHOR

=====

Written for the R65 system.
Implementation by R. Richarz and ChatGPT.

END

===

End of NR0FF manual. □

□

paint

`PAINT filename[.cyclus][,drive]`

Painting on the R65 low resolution graphics canvas

arrows	move cursor
C	clear canvas
M	switch color
p	paint a dot at cursor position
L	drawing line mode
R	drawing rectangle mode
S	drawing character mode
return	draw object and exit mode
esc	exit mode without drawing
W	write canvas to disk
Q	write canvas to disk, quit
K	kill program without writing

pcodes

PCODES filename[disk,from,to]

Display P-codes of a Pascal Program.
Pascal source code and object code must be
on the same drive.

If the program has been compiled with the /L
option (line numbers), pcode displays the
source line together with the pcodes for this
line, and pcodes of a specified range of
lines (from/to) can be displayed. Use SETB
and RESETB to set breakpoints.

pedit

PEDIT filename

ESC	Enter command
t	Go to top of file
b	Go to bottom of file
l n	Go to line n
f	Find
a	Find again
z	Clear marks
c n	Mark lines of copy
p	Paste copied lines
m	Move copied lines
d n	Delete n lines
w	Write file
q	Write file and quit
k	Kill (quit without writing)
?,h	Help

plotlib

LIBRARY PLOTLIB;

CONSTANTS

```
XSIZE=223;  
YSIZE=117;  
XWORDS=28;  
WHITE=0;  
INVERSE=1;  
BLACK=2;  
PLOTDEV=@128;
```

MEMORY

```
KEYPRESSED=$1785: char&;
```

VARIABLES

```
_xcursor, _ycursor: integer;
```

ROUTINES

```
proc _delay10msec(time:integer);  
func _syncscreen;  
proc _grinit;  
proc _grend;  
proc _cleargr;  
proc _fullview;  
proc _splitview;  
proc _plot(x,y,c:integer);  
proc _move(x,y:integer);  
proc _draw(x,y,c:integer);  
proc _plotmap(x,y,m:integer);  
proc _waitforKEY;
```

R65

R65 QUICK REFERENCE

=====

MONITOR COMMANDS

=====

DIR Directory listing.
COPY Copy file between drives.
DELETE Delete a file.
EDIT Edit text file.
LOAD Load file to memory.
STORE Store memory sektion to disk.
GO Execute machine code in memory.
DIS Disassemble memory.
REG Show CPU registers.
DATE Show or set date.
FLOPPY Select disk image.
IMPORT Import host file.
EXPORT Export file to host.
RUN Run a maschine code program.

PASCAL

=====

COMPILE Compile Pascal program.
EDIT Edit a text file.
name Execute Pascal program name.
LIB Load Pascal library.

NR0FF

=====

NR0FF Format text document.
.TH Manual header.
.SH Section heading.
.SS Subsection heading.
.PP New paragraph.
.IP Indented paragraph.
.nf / .fi No fill / fill mode.
.ta Set tab stops.

SCREEN CONTROL

=====

CT-E Clear line.
CT-RTN Clear to end of display.

CT-RGT	Insert character.
CT-LFT	Delete character.
CTRL-A	Cursor home.
CT-UP	Roll up.
CT-DN	Roll down.

ralib

LIBRARY RALIB;

CONSTANTS

```
FREAD=$00;  
FWRITE=$20;  
FNEW=$30;
```

ROUTINES

```
func _uppercase(ch:char):char;  
func _attach(fname:array[15] of char;  
    cyclus,drive,operation,size,start:integer;  
    subtype:char):file;  
func _getsize:integer;  
proc _getword(device:file; address:integer;  
    var word:integer);  
proc _putword(device:file; address:integer;  
    word:integer);  
proc _getreal(device:file; address:integer;  
    var rvalue:array[1] of %integer);  
proc _putreal(device:file; address:integer;  
    rvalue:array[1] of %integer);
```


striolib

LIBRARY STRIOLIB;

CONSTANTS

 NAMESIZE = 15;

MEMORY

 filerr = \$00db: integer&;

VARIABLES

 scarg: integer;

ROUTINES

proc _argerror(e: integer);

proc _checkfilerr;

proc _sgetval(var value: integer;
 var default: boolean);

proc _sgetstring(string: cpnt; var default: boolean;
 var cyclus, drive: integer);

proc _ssetsubtype(sname: cpnt; subtype: char;
 force: boolean);

proc _strfio(name: cpnt; drive: integer);

proc _getdiskname(s: cpnt; drive: integer);

proc _change_disk(s: cpnt; drv: integer);

proc srunprog(name: cpnt; cyc: integer; drv: integer);

strlib

LIBRARY STRLIB;

CONSTANTS

STRSIZE=64;
ENDMARK=chr(0);

ROUTINES

```
proc _runerr(e:integer);  
func _new: cpnt;  
proc _release(s: cpnt);  
func _strlen(strin:cpnt):integer;  
proc _strcpy(strin, strout:cpnt);  
proc _stradd(strin, strinout:cpnt);  
func _strcmp(s1,s2:cpnt):integer;  
func _strpos(ch:char; s1:cpnt;  
             start:integer): integer;  
func _strread(f:file; s:cpnt;  
              var ateof0:boolean):integer;  
proc _hexstr(d:integer; s:cpnt);
```

syslib

LIBRARY SYSLIB;

CONSTANTS

```
TAB8      = chr(9);
HOM        = chr(1);
CSC        = chr($11);
LF         = chr($a);
FF         = chr($c);
CR         = chr($d);
EOF        = chr($7f);
NORVID     = chr($0b);
INVVID     = chr($0e);
PRTON      = chr($12);
PRTOFF     = chr($14);
MMAXSEQ    = 8;
TOPMEM     = $c780;
MAXINT     = $7fff;
INPUT      = @0;
OUTPUT     = @0;
KEY        = @1;
PRINTER    = @1;
```

MEMORY

```
RUNERR     = $000c: integer&;
ENDSTK     = $000e: integer;
BUFFPN     = $0015: integer&;
IOCHECK    = $0023: boolean&;
CURPOS     = $00ee: integer&;
```

VARIABLES

```
_day:      integer;
_month:    integer;
_year:     integer;
```

ROUTINES

```
proc _setemucom(i:integer);
func _getbcd(address: integer): integer;
proc _getdate;
proc _prtdate(device: file);
func _abs(x: integer): integer;
func _mod(x,n: integer): integer;
proc _tab(pos: integer);
proc _abort;
```

```
proc _prtext8(device: file;  
              text: array[7] of char);  
proc _prtext16(device: file;  
               text: array[15] of char);  
func _random: integer;
```

table

graph

display the graph of function values
stored in table

table

make a table for display with graph

teklib

LIBRARY TEKLIB;

CONSTANTS

```
MAXX = 1023;
MAXY = 780;
MAXCOLUMNS = 74;
MAXLINES    = 35;
SOLID       = 1;
DOTTED      = 2;
DOTDASH     = 3;
SHORTDASH   = 4;
LONGDASH    = 5;
PLOTTER     = @1;
```

VARIABLES

```
_xs,_ys: integer;
```

ROUTINES

```
proc _clearscreen;
proc _starttek;
proc _endtek;
proc _startdraw(x1,y1:integer);
proc _draw(x2,y2:integer);
proc _enddraw;
proc _drawvector(x1,y1,x2,y2:integer);
proc drawrectangle(x1,y1,x2,y2:integer);
proc _moveto(x1,y1: integer);
proc _delay10msec(time:integer);
proc _setchsize(size:integer);
proc _setlinemode(type:integer);
```

timelib

LIBRARY TIMELIB;

VARIABLES

```
    tenmillis,seconds,minutes,hours,  
    difftenmillis: integer;
```

ROUTINES

```
proc gettime;  
proc prttime(device: file);  
func timediff: real;
```

view

VIEW filename

ESC, q	Quit
CUP, CDOWN	Scroll up/down
CLEFT, CRIGHT	Scroll 12 lines up/down
/, f	find string
n	next hit
?, h	help

wildlib

LIBRARY WILDLIB;

CONSTANTS

```
    NAMESIZE    = 15;  
    NUMENTRIES  = 255;
```

MEMORY

```
    FILNAM=$0301: array[NAMESIZE] of char&;
```

ROUTINES

```
proc _test(s1:array[NAMESIZE] of char;  
    var found:boolean);  
proc _findentry(var nm:array[NAMESIZE] of char;  
    drv:integer;var ent: integer;  
    var fnd,lst:boolean);  
proc _sfindentry(name: cpnt; drv:integer;  
    var ent: integer;  
    var fnd,lst: boolean);  
proc _writename(nm1:array[15] of char);
```